

# Optimal Planning of Emergency Evacuations for Electric Vehicles Leveraging Equivalent Circuit Models

## Authors

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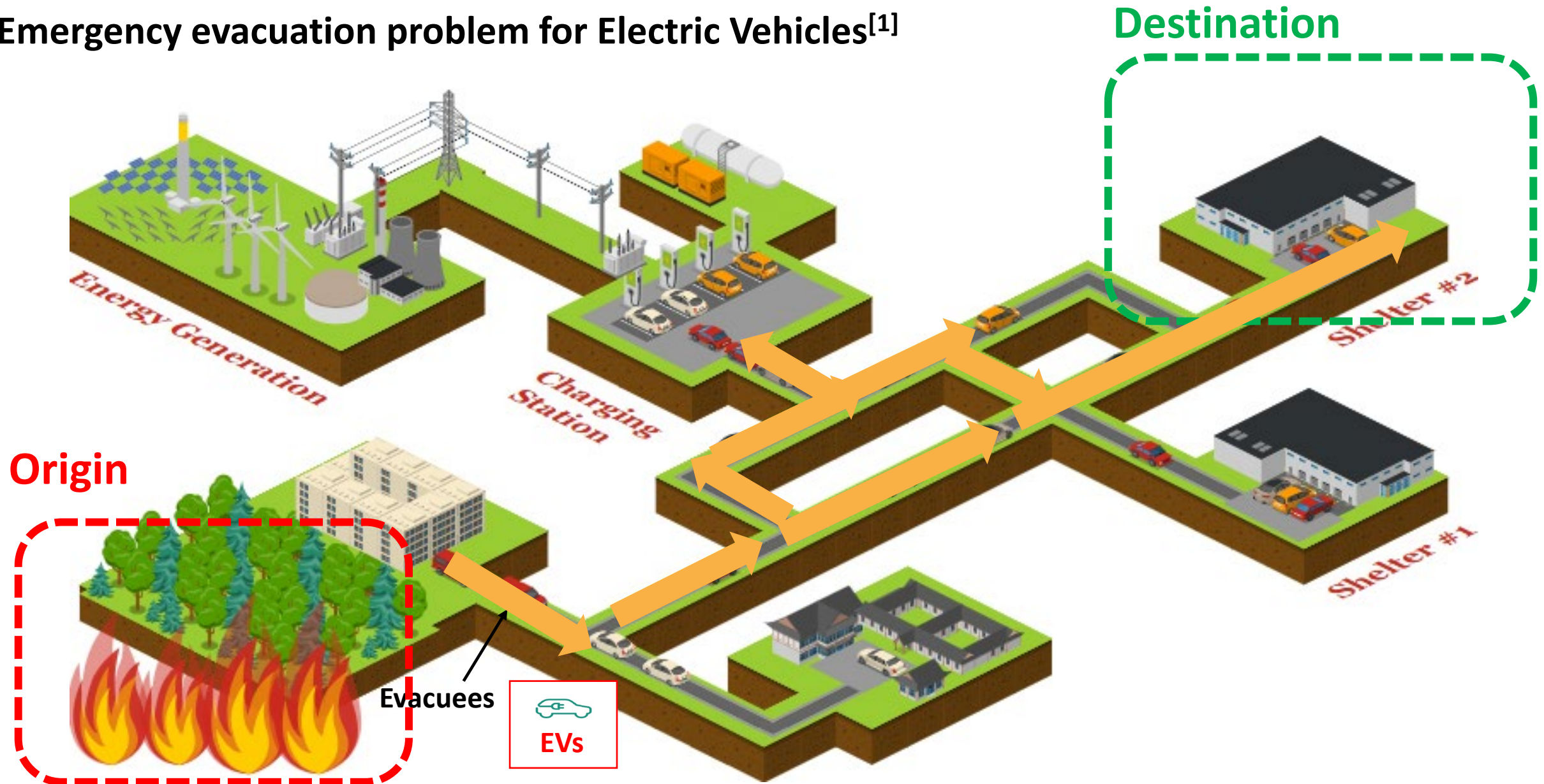
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University of California, Berkeley<sup>c</sup>



# Emergency evacuation problem for Electric Vehicles<sup>[1]</sup>

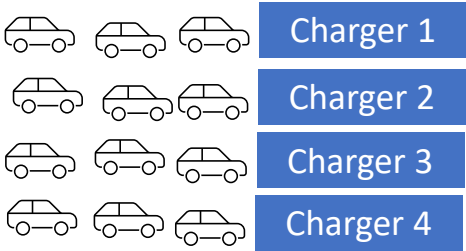


# Emergency evacuation problem for Electric Vehicles<sup>[1]</sup>

## Power Outage



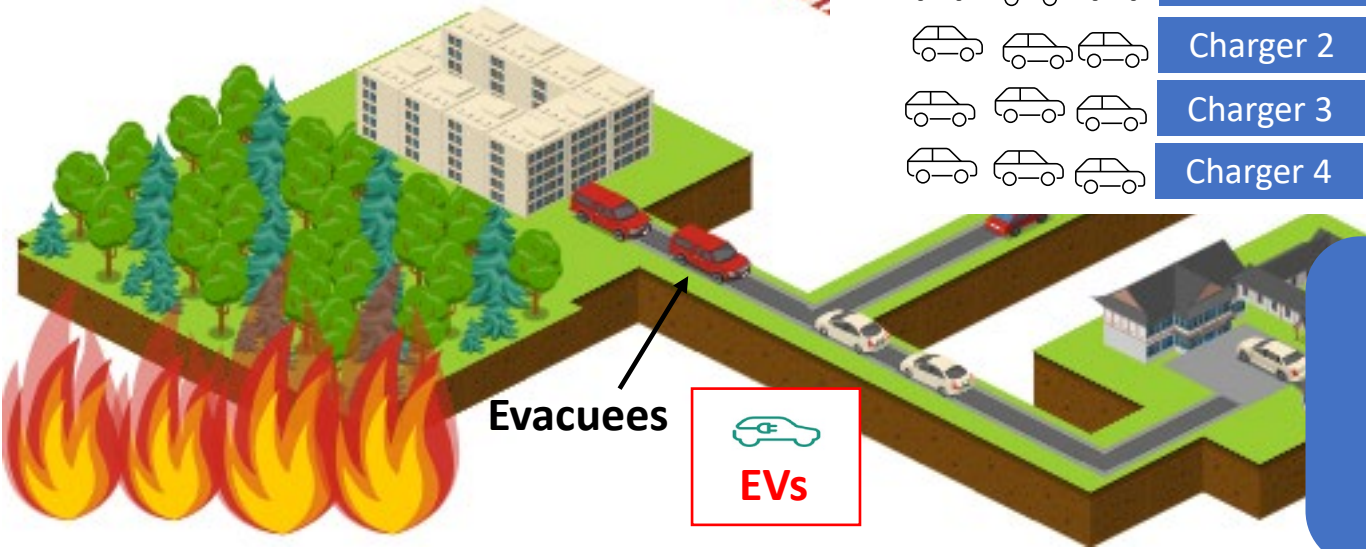
## Charging Congestion



## Traffic Congestion



Lake Toahoe, Caldor Fire Source: Josh Edelson, AFP

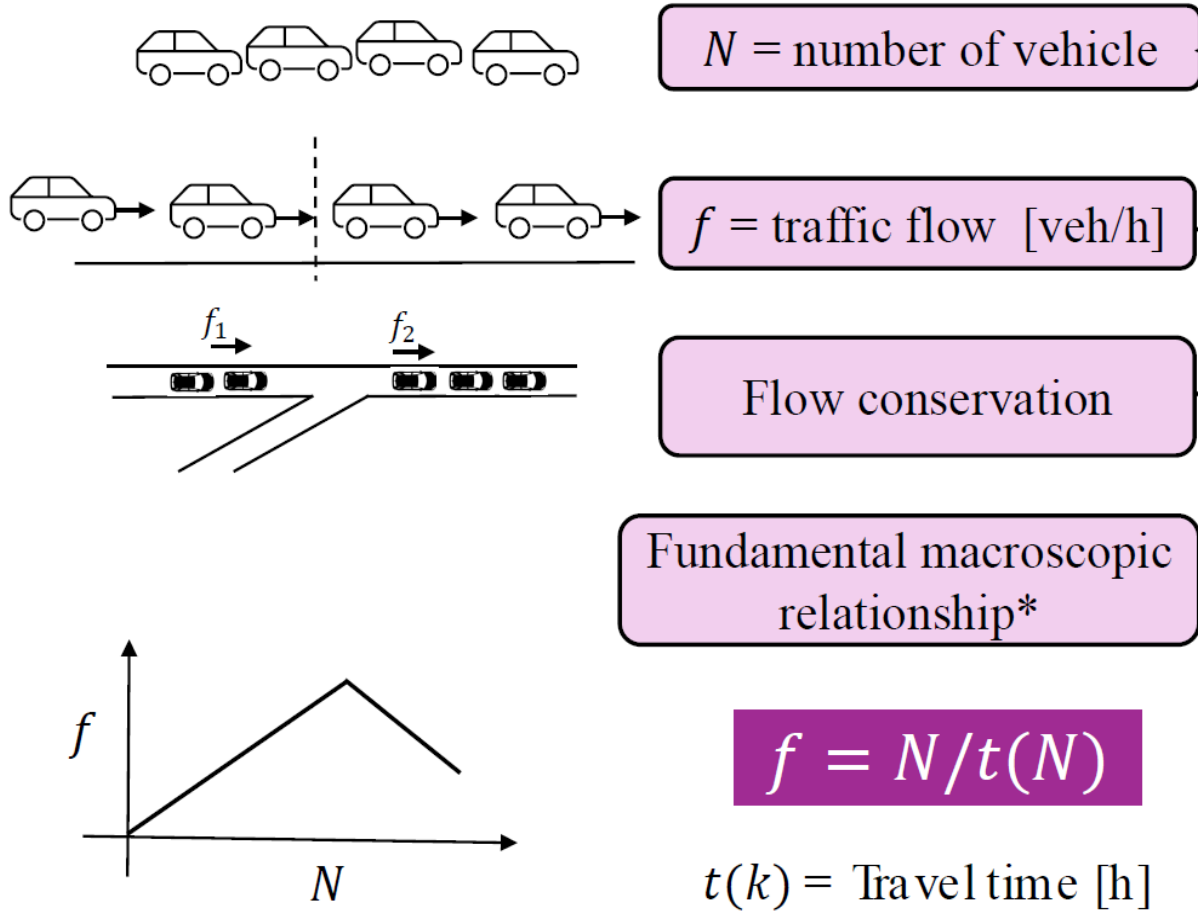


Need of optimal vehicle routing and charging algorithm to alleviate these challenges

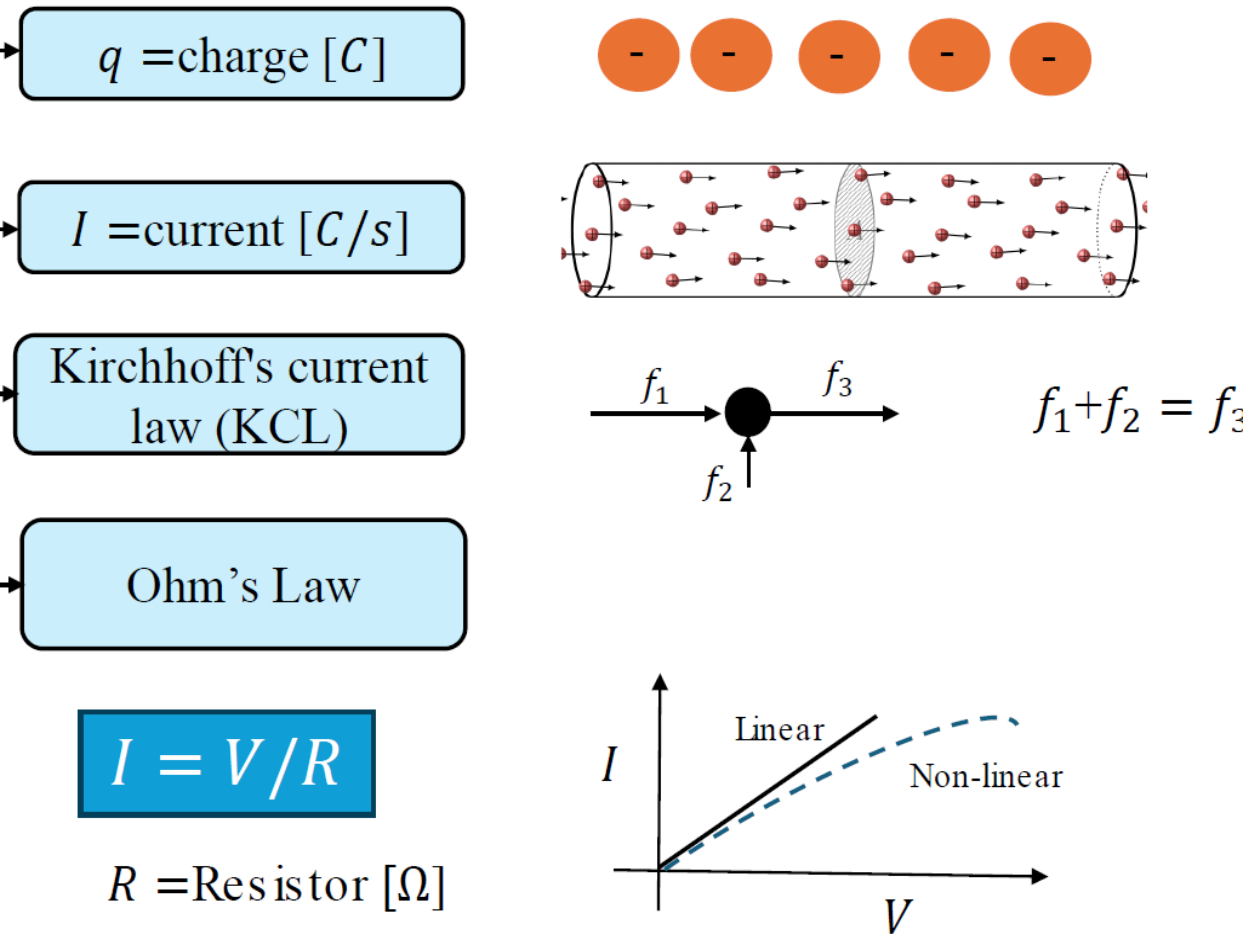
1 [1] Bell, Michael GH, and Yasunori Iida. *Transportation network analysis*. 1997.

# Equivalent Circuit Model (ECM) for Transportation Network<sup>[2]</sup>

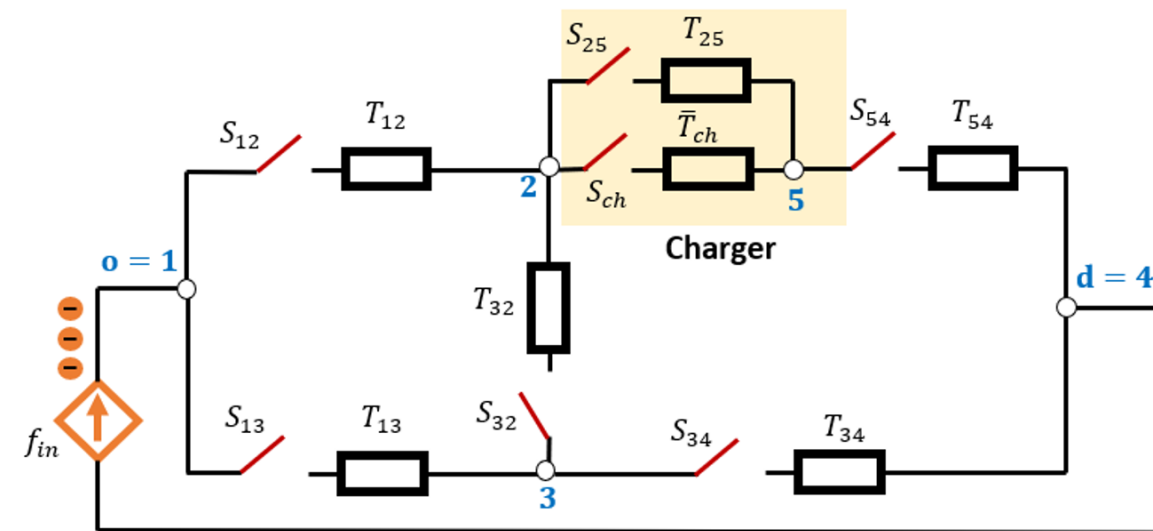
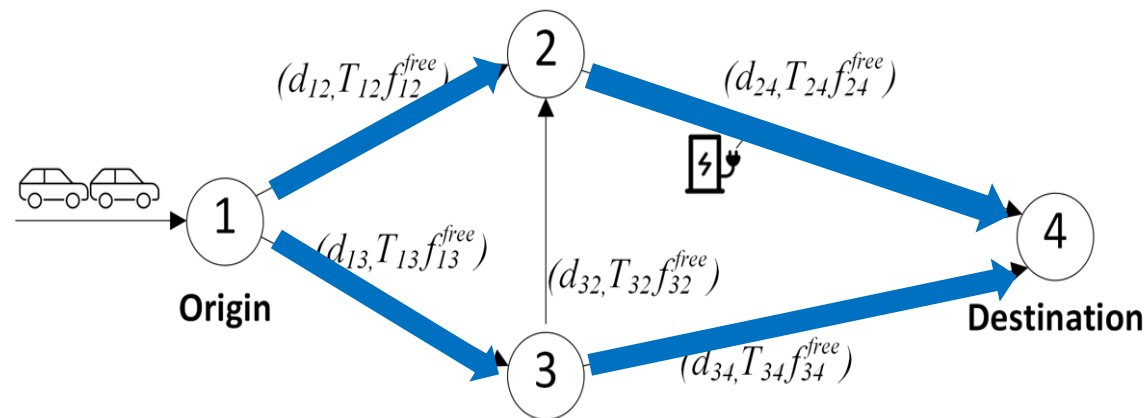
## Macroscopic Traffic Flow



## Equivalent Circuit Model (ECM)

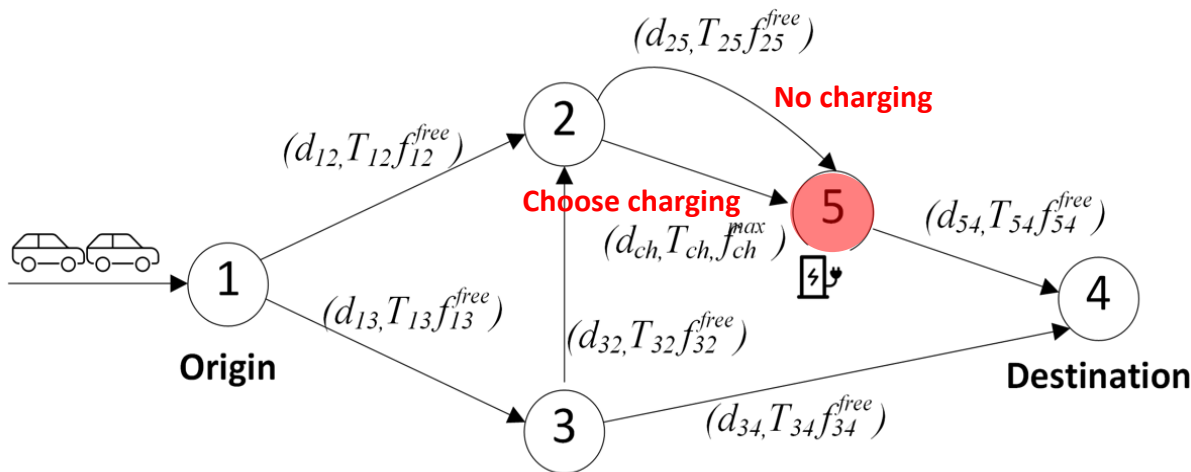


# Equivalent Circuit Model (ECM) for Transportation Network



ECM framework

Decision variables are  $S_{ij}$  and  $S_{ch}$  which take Boolean values  $\{0,1\}$

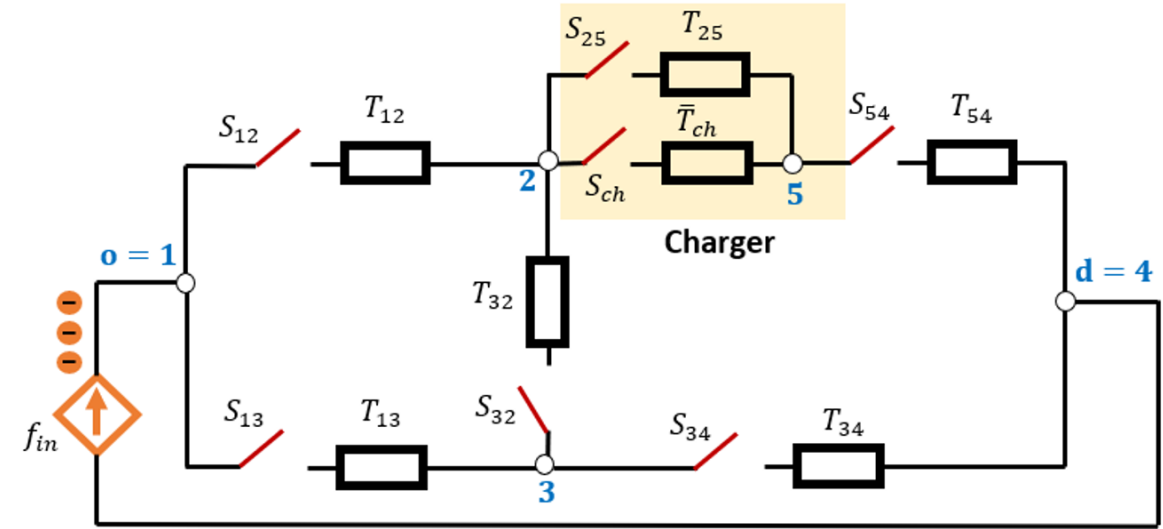


# Integer Linear program (ILP) for Emergency Evacuation

$$\begin{aligned}
 (\text{node 1}) : \quad & I_{12} + I_{13} = I_{in} \\
 (\text{node 2}) : \quad & I_{12} + I_{32} - I_{25} - I_{ch} = 0 \\
 (\text{node 3}) : \quad & I_{13} - I_{32} - I_{34} = 0 \\
 (\text{node 4}) : \quad & I_{54} + I_{34} = I_{in} \\
 (\text{node 5}) : \quad & I_{ch} + I_{25} - I_{54} = 0
 \end{aligned}$$



$$\mathbf{A}_{KCL} \mathbf{S} + \mathbf{A}_{KCL}^{ch} \mathbf{S}^{ch} = \mathbf{B}_{KCL}$$



ECM framework

$$\min_{\mathbf{S}, \mathbf{S}^{ch}} \mathbf{T}^\top \mathbf{S} + \mathbf{T}_{ch}^\top \mathbf{S}^{ch}$$

Total evacuation time

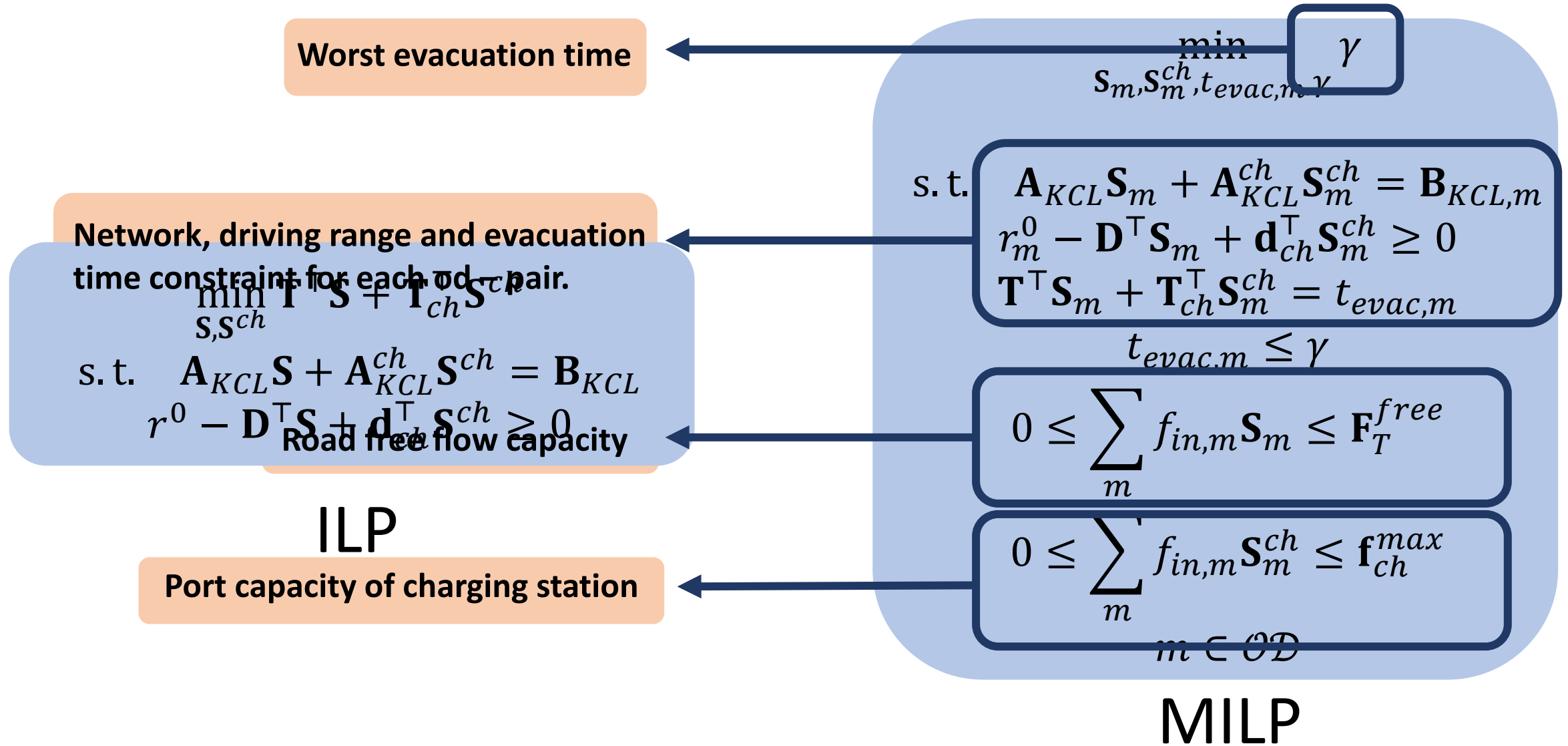
s. t.  $\mathbf{A}_{KCL} \mathbf{S} + \mathbf{A}_{KCL}^{ch} \mathbf{S}^{ch} = \mathbf{B}_{KCL}$

Network constraints

$$r^0 - \mathbf{D}^\top \mathbf{S} + \mathbf{d}_{ch}^\top \mathbf{S}^{ch} \geq 0$$

Driving range constraint

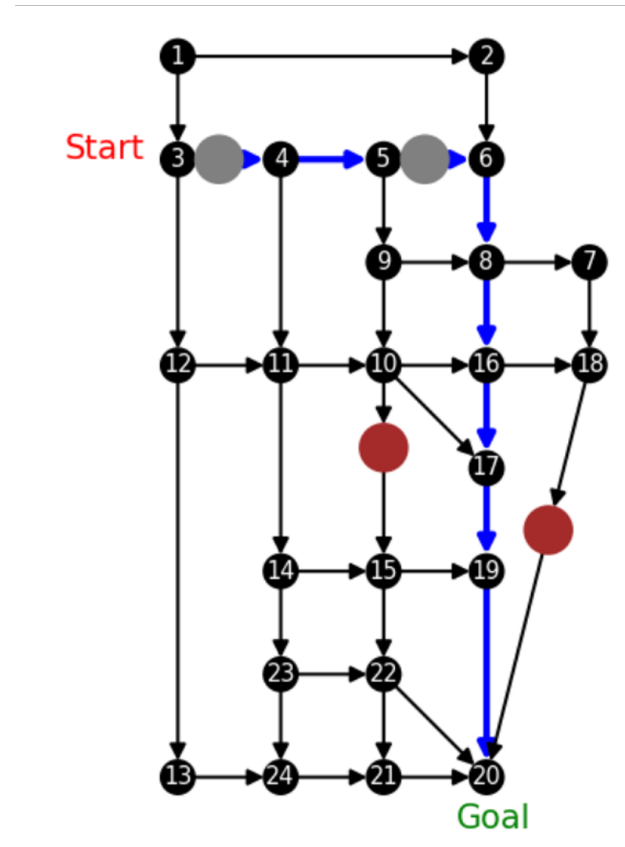
# Mixed Integer Linear program (MILP) for multiple (od)-pairs



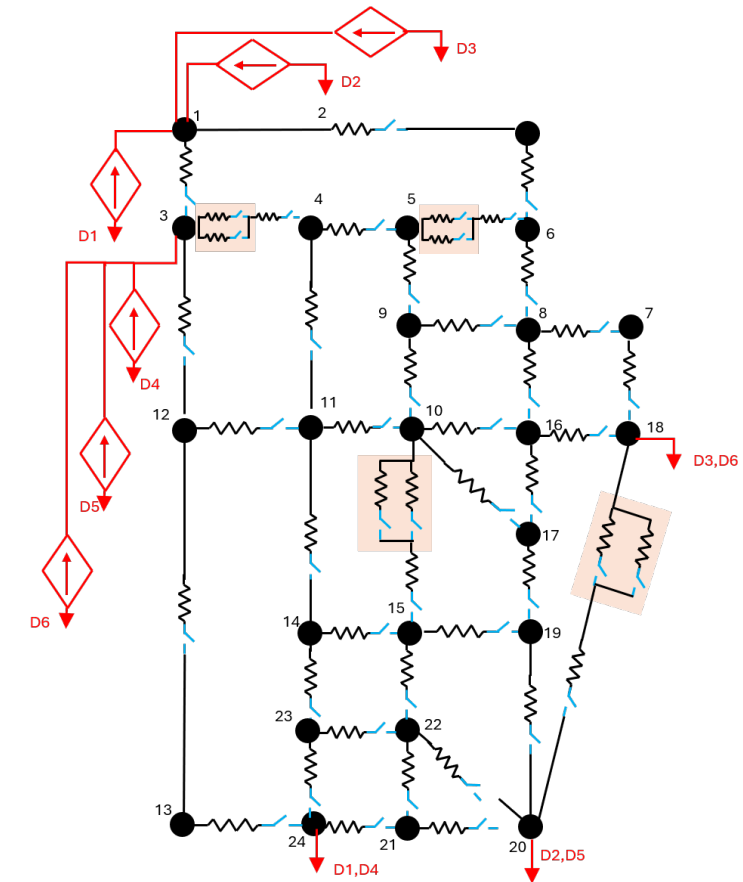
# Sioux Falls Transportation Network<sup>[3]</sup>

## Parameters

- Number of nodes: 24
- Number of links: 38
- Number of (od)-pairs: 6
- Charging speed: 20 kW
- $f_{ij}^{free}$ : 1000 vehicles/hr
- $f_{in}$ : 50 vehicles/hr

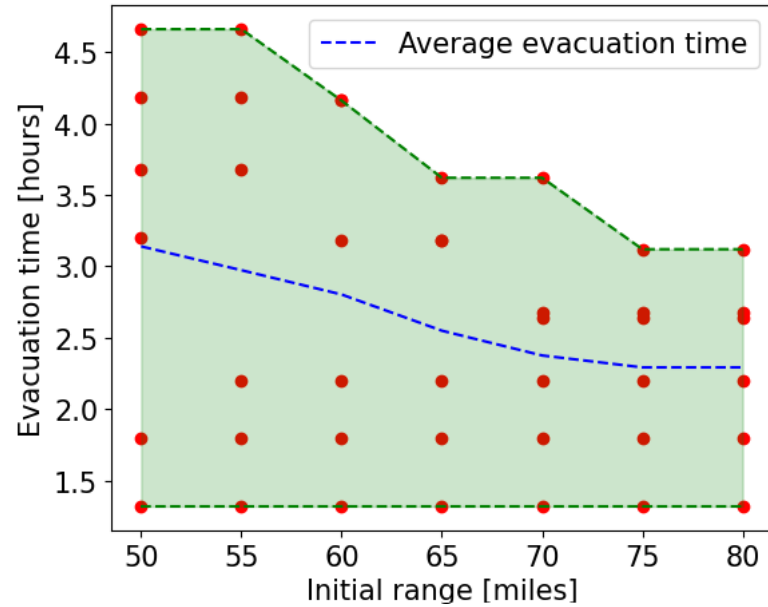


Sioux falls network

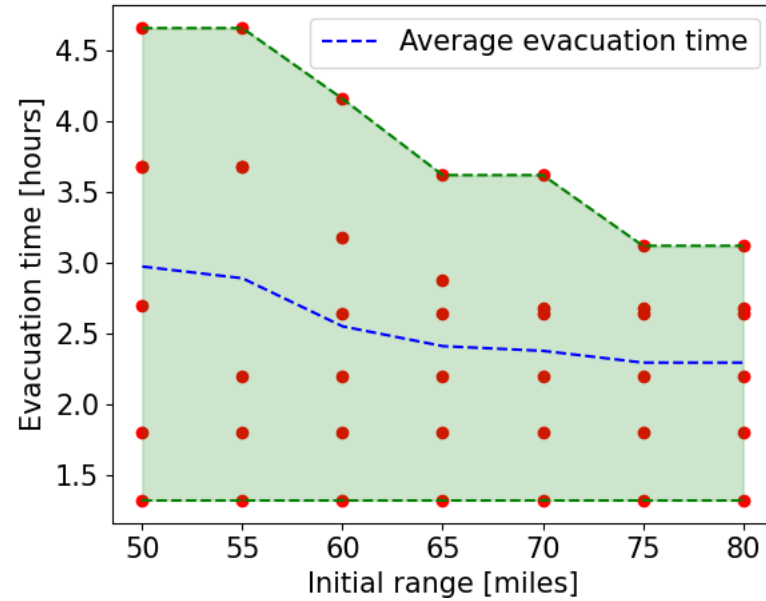


ECM framework

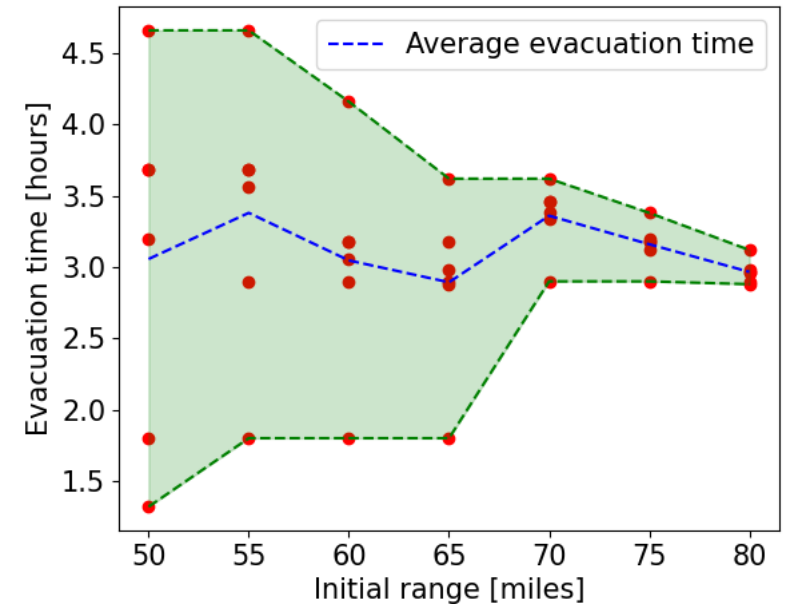
# Sensitivity Analysis with different cost metrics



Minimizing worst evacuation time

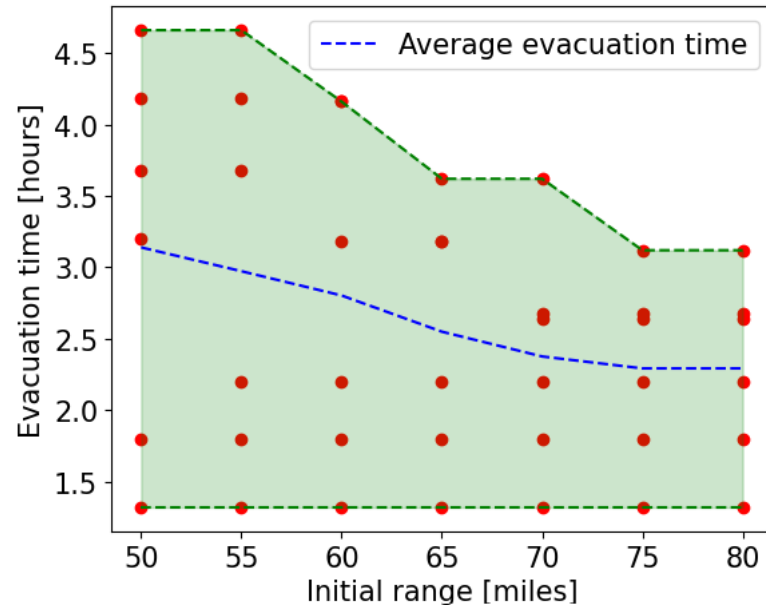


Minimizing average evacuation time



Minimizing deviation between all evacuation time

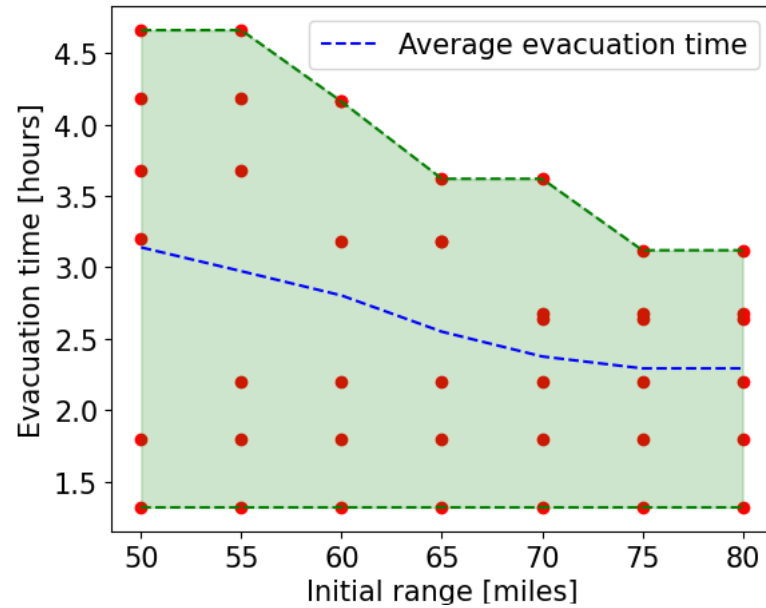
# Sensitivity Analysis with different cost metrics



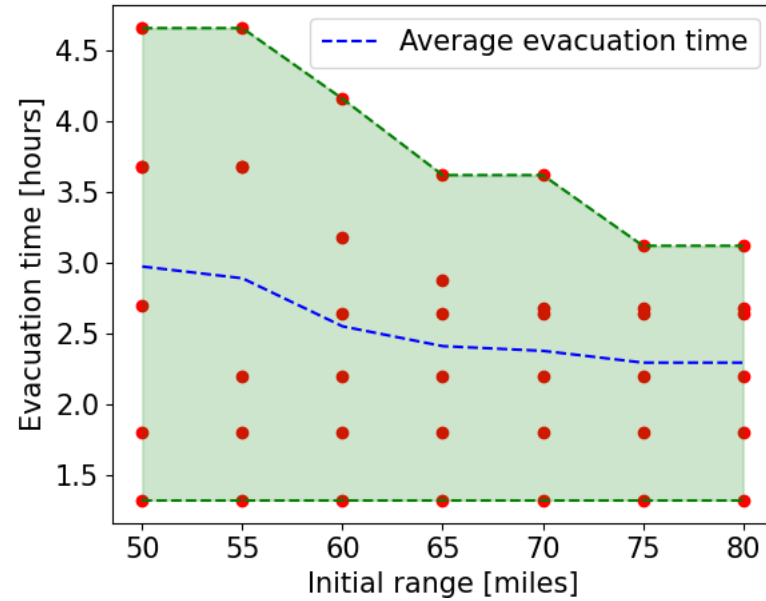
- Worst evacuation time decreases as we increase initial range.
- Electric vehicles with sufficient initial range don't need to stop at charging station.

Minimizing worst evacuation time

# Sensitivity Analysis with different cost metrics



Minimizing worst evacuation time

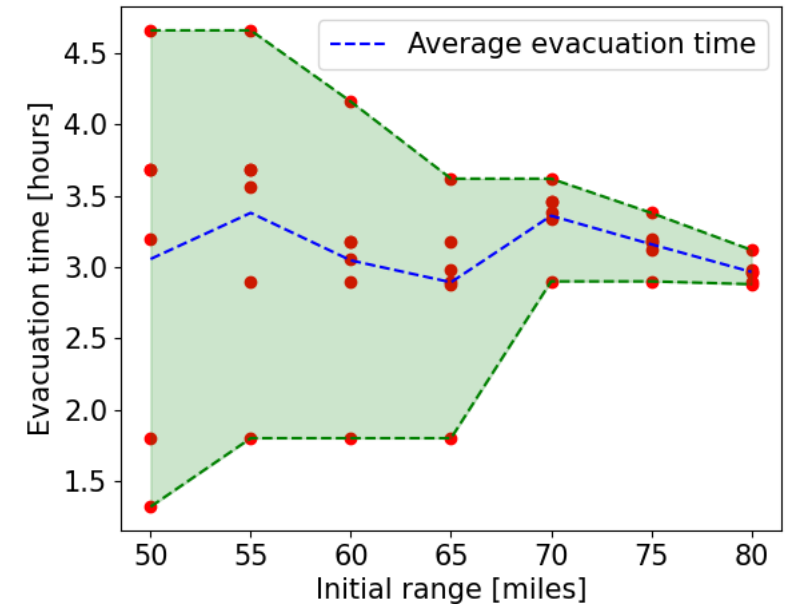


Minimizing average evacuation time

Average evacuation time (shown by **dash blue line**) starts at **3.2 hours** in left plot whereas the value starts at **3 hours** in right plot.

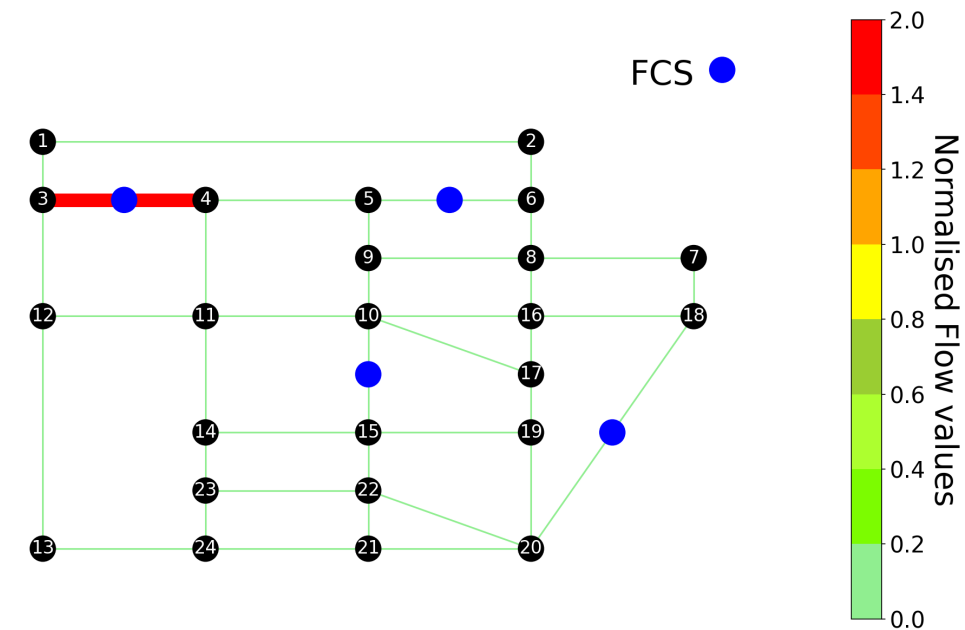
# Sensitivity Analysis with different cost metrics

- Deviation in the evacuation times of the six od - pairs decreases as we increase the initial range.
- The optimization enforces longer routes for certain OD pairs to reduce evacuation time deviations, resulting in a higher average evacuation time compared to the other two metrics.

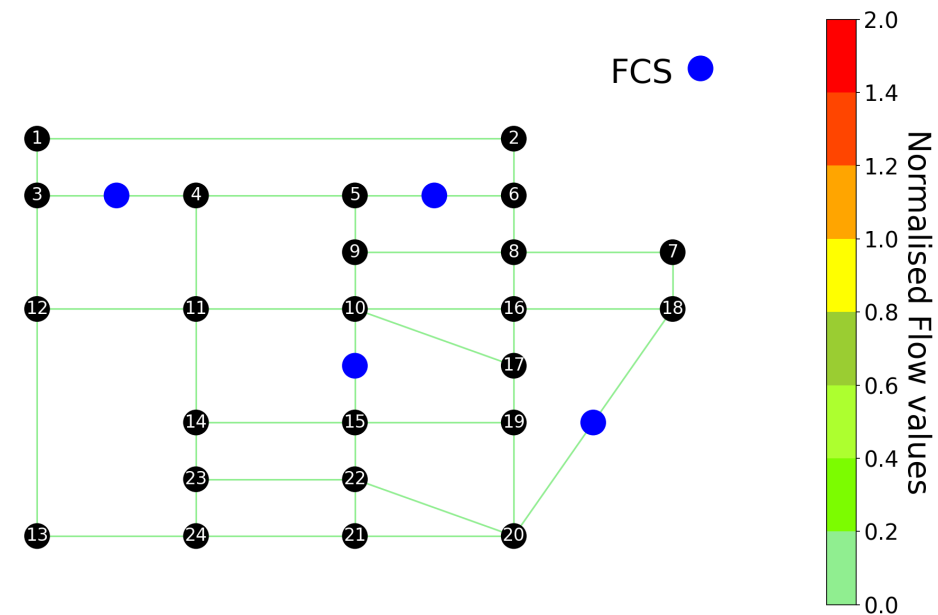


**Minimizing deviation between all evacuation time**

# Comparison with shortest route algorithm

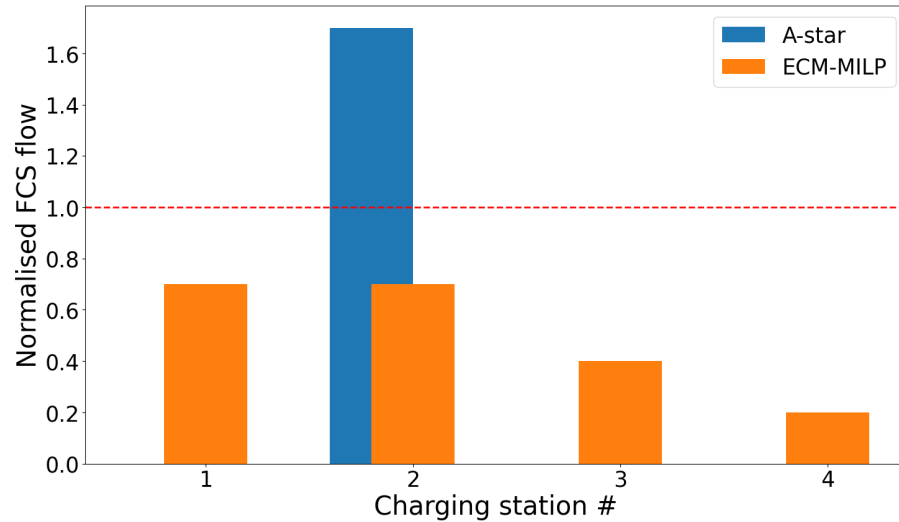


Expected road traffic flow for different links for shortest route evacuation plan (  $A^*$  algorithm).



Expected road traffic flow for different links for ECM-based optimization evacuation plan.

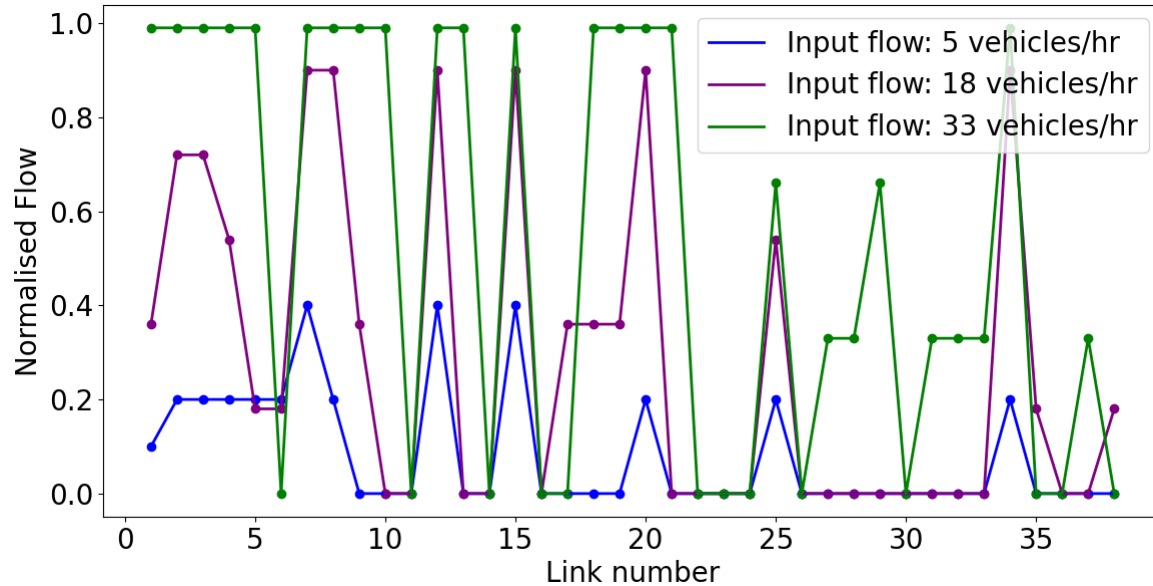
# Comparison with shortest route algorithm



Normalized flow rate of different FCS in  
Sioux Falls map

- The shortest-route evacuation plan prioritizes **short-term rewards**, directing all EVs to the nearest charging station (#2), which leads to **traffic congestion and queuing delays**.
- The ECM-based optimization adopts a global perspective, distributing EVs across all available charging stations to mitigate traffic congestion and queuing delays.

# Sensitivity Analysis of road capacity of different links



- Road flow capacity increases as we increase input flow of od - pairs.
- As the input flow increases, a **spillover effect** occurs—some road links reach their free-flow limits, causing excess traffic to be diverted to previously underused links.

The normalized flow values  $(f_{in}/f_{ij}^{free})$  of different edges for the Sioux Falls network for various values of  $f_{in}$ . Here,  $f_{ij}^{free} = 100$  vehicles/hr.

# Conclusions

- Proposed a novel **equivalent circuit model** to formulate the emergency evacuation problem.
- MILP framework provides **routing and recharging strategies** that minimize overall evacuation time while satisfying flow constraints.
- **Numerical simulations** based on Sioux Falls demonstrate the effectiveness of the proposed approach under various evacuation scenarios.

# Future Work

- Integration of the ECM with **mobile chargers** and real-time evacuation planning.
- Incorporating **Bureau of Public Roads (BPR)** function into the ECM model-based optimization.
- Converting the charging station stoppage time from a parameter to a decision variable.

# Thank you

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Google Scholar  
Link

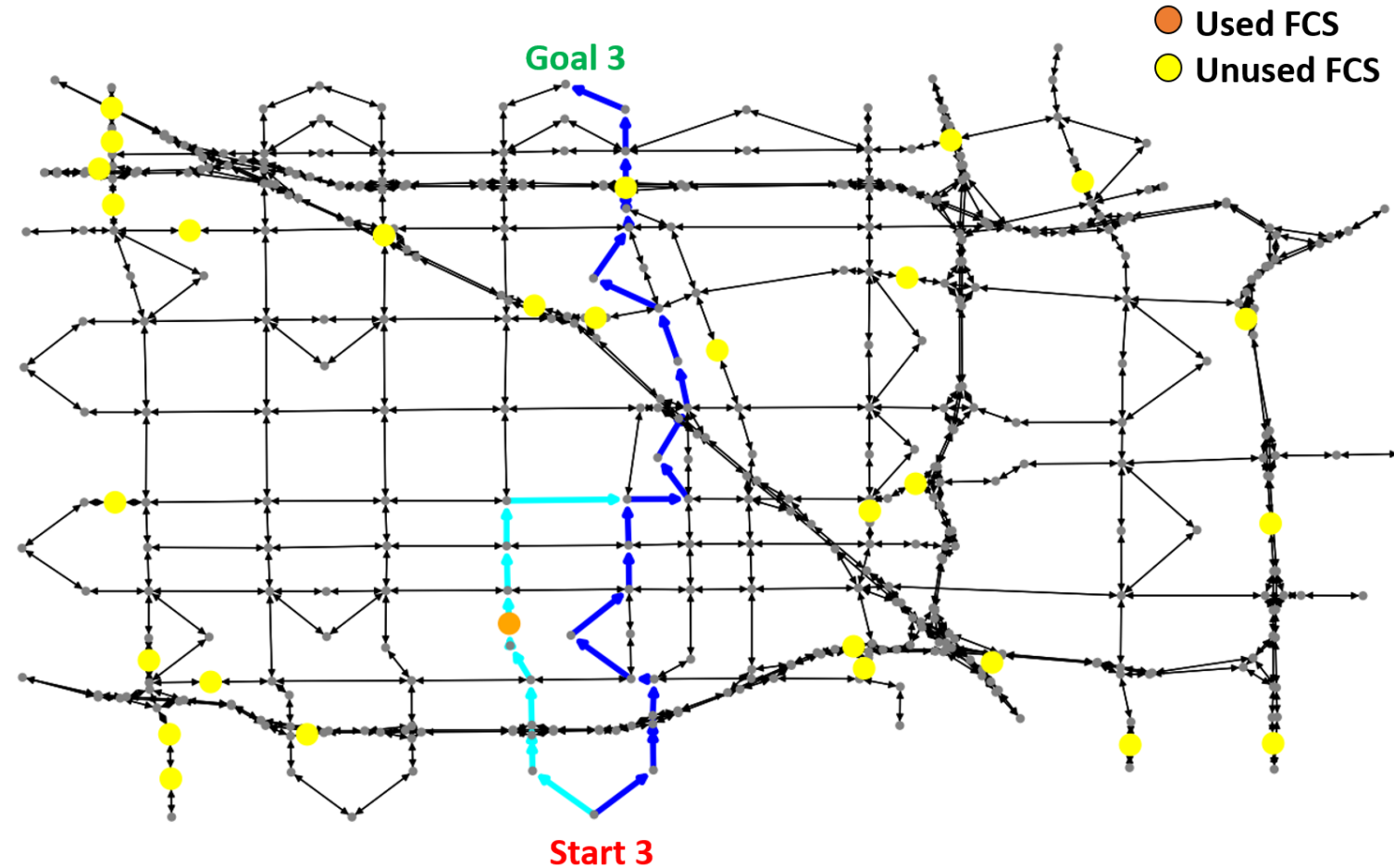


LinkedIn Link

# Anaheim Transportation Network<sup>[2]</sup>

## Parameters

- Number of nodes: 416
- Number of links: 914
- Number of (od)-pairs: 6
- Charging speed: 50 kW
- $f_{ij}^{free}$ : 1800-12600 vehicles/hr
- $f_{in}$ : 100 vehicles/hr



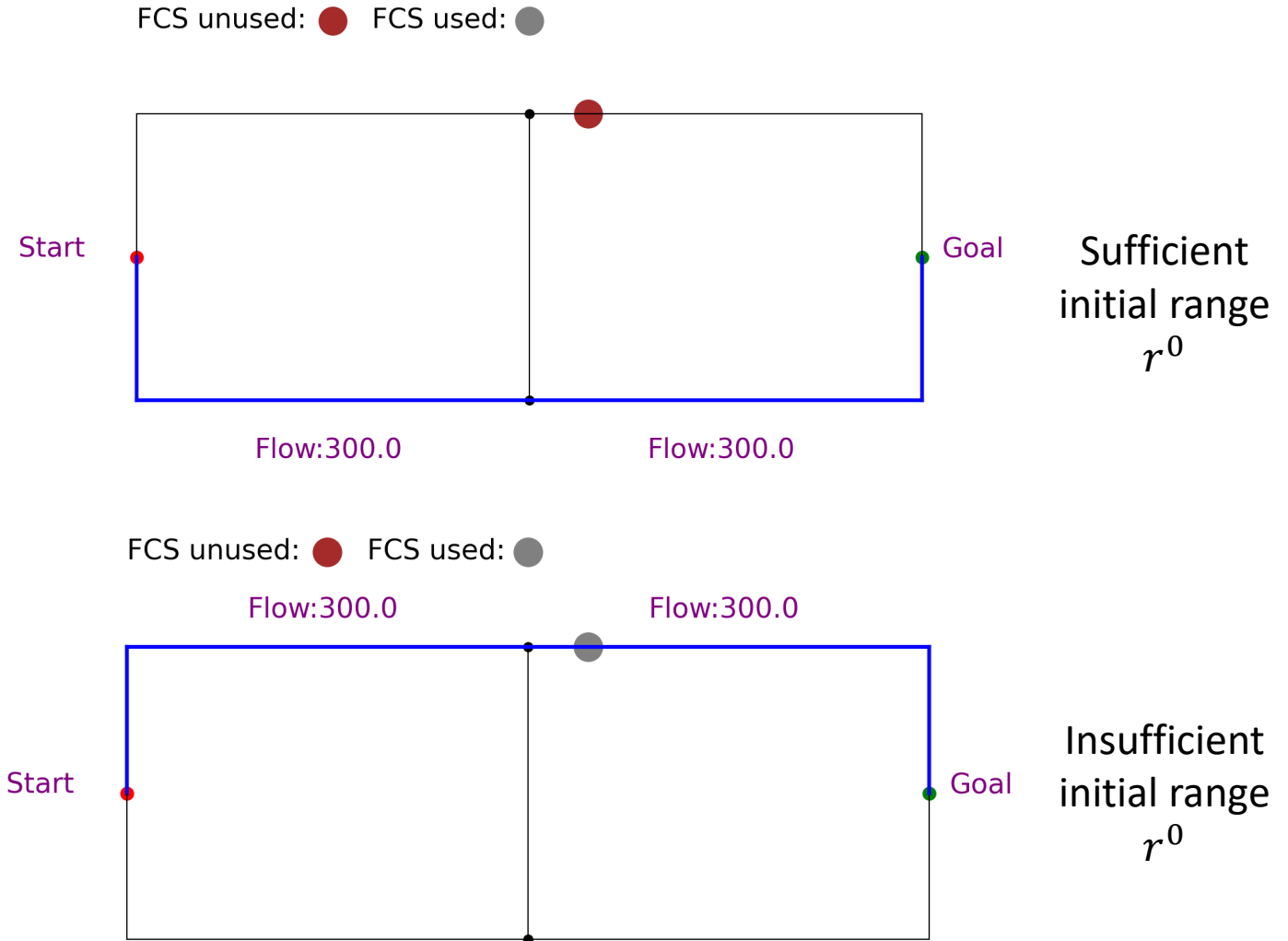
[2] Transportation networks for research core team. Technical report, Transportation Networks for Research. URL <https://github.com/bstabler/TransportationNetworks>.

## Computation time comparison

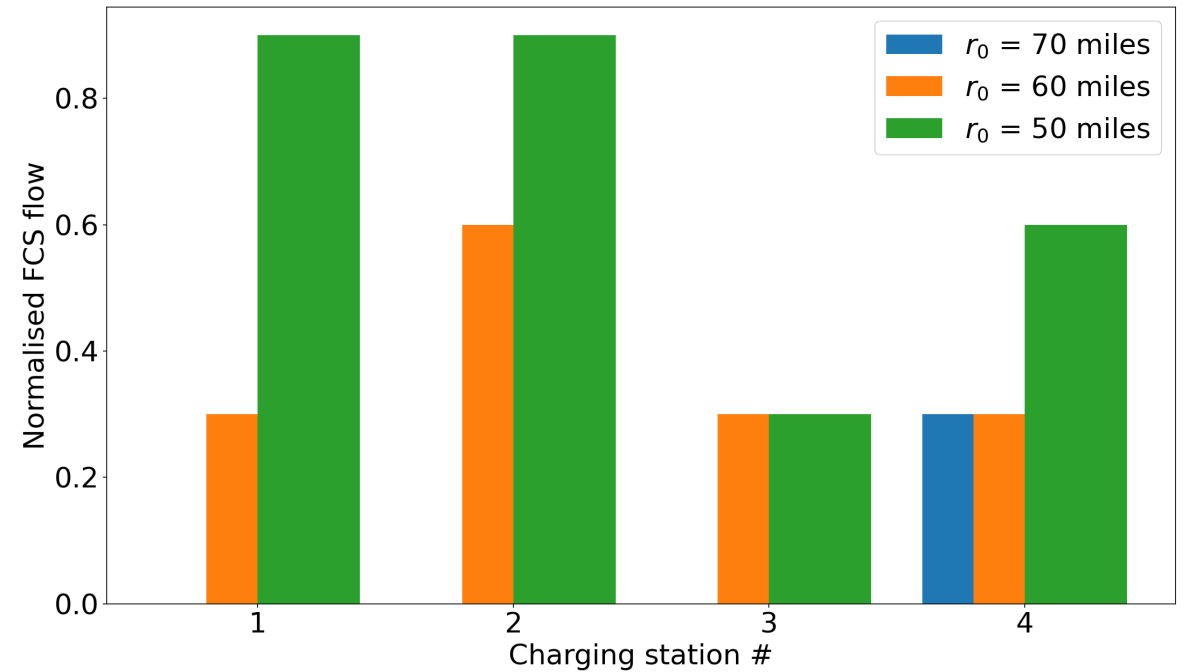
| # OD-pairs | Sioux Falls        | Anaheim              |
|------------|--------------------|----------------------|
|            | 24 nodes, 38 links | 416 nodes, 914 links |
| 2          | 0.32 seconds       | 30.75 seconds        |
| 4          | 0.34 seconds       | 55.184 seconds       |
| 8          | 0.56 seconds       | 1.85 minutes         |
| 10         | 1.2 seconds        | 2.01 minutes         |
| 15         | 2.2 seconds        | 1.42 hours           |
| 20         | 3.7 seconds        | > 5 hours            |

# Integer Linear program (ILP) for Emergency Evacuation

$$\begin{aligned} & \min_{\mathbf{S}, \mathbf{S}^{ch}} \mathbf{T}^\top \mathbf{S} + \mathbf{T}_{ch}^\top \mathbf{S}^{ch} \\ \text{s. t. } & \mathbf{A}_{KCL} \mathbf{S} + \mathbf{A}_{KCL}^{ch} \mathbf{S}^{ch} = \mathbf{B}_{KCL} \\ & r^0 - \mathbf{D}^\top \mathbf{S} + \mathbf{d}_{ch}^\top \mathbf{S}^{ch} \geq 0 \end{aligned}$$



# Sensitivity Analysis of port capacity of different charging locations



The normalized flow values ( $f_{in}/f_{ch,(i,j)}^{max}$ ) of fixed charging stations (FCS) for the Sioux Falls network for various values of  $r^0$ .